Eric Xing

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Education

Washington University in St. Louis St. Louis, MO
*Ph.D. in Computer Science; **GPA: 4.00*** *Aug 2023 – Present*

Western Kentucky University Bowling Green, KY
*B.S. in Computer Science with Minor in Mathematics, Summa Cum Laude; **GPA: 4.00*** *May 2019 – Aug 2023*

Under Review

[1] **Xing, E.**, Li, V., Cher, D., & Jacobs, N. *QuASI: Query-Adaptive Sparse Indexing for Multimodal Search.* Annual AAAI Conference on Artificial Intelligence (**AAAI'27** Submission).

[2] Cher, D., **Xing, E.**, Li, K., Wei, B., Corley, I., Jacobs, N. *Chronosphere: Space-Time Tessellation of Local Climate Experts.* Annual AAAI Conference on Artificial Intelligence (**AAAI'27** Submission).

[3] **Xing, E.**, Wei, B., Cher, D., & Jacobs, N. *Learning to Allocate Tokens for Vision Transformers.* Conference on Neural Information Processing Systems (**NeurIPS'26** Submission.)

[4] Cher, D., Wei, B., Sastry, S., **Xing, E.**, Jarosz, P., Sandvig, R., & Jacobs, N. *Phenological Plant Embeddings.* Conference on Neural Information Processing Systems (**NeurIPS'26** Submission).

Publications

[1] Wei, B., Sastry, S., Cher, D., **Xing, E.**, & Jacobs, N. *GeoDiT-Ω: Unified Spatial Control for Satellite Image Synthesis with Any Geospatial Primitive.* European Conference on Computer Vision (Accepted to **ECCV'26**).

[2] Cher, D., Iqbal, H., **Xing, E.**, Wei, B., & Jacobs, N. *Tessellating The Earth.* European Conference on Computer Vision (Accepted to **ECCV'26**).

[3] **Xing, E.**, Stylianou, A., Pless, R., & Jacobs, N. *QuARI: Query Adaptive Retrieval Improvement.* The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS'25**). [link]

[4] Sastry, S., Dhakal, A., **Xing, E.**, Khanal, S. & Jacobs, N. *Global and Local Entailment Learning for Natural World Imagery.* The International Conference on Computer Vision 2025 (**ICCV'25**). [link]

[5] Qiao, F., Xiong, Z., **Xing, E.**, & Jacobs, N. *Towards Open-World Generation of Stereo Images and Unsupervised Matching.* The International Conference on Computer Vision 2025 (**ICCV'25**). [link]

[6] **Xing, E.**, Kolouju, P., Pless, R., Stylianou, A. & Jacobs, N. *ConText-CIR: Learning from Concepts in Text for Composed Image Retrieval.* The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (**CVPR'25**). [link]

[7] Dhakal, A., Sastry, S., Khanal, S., Ahmad, A., **Xing, E.**, & Jacobs, N. *RANGE: Retrieval Augmented Neural Fields for Multi-Resolution Geo-Embeddings.* The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (**CVPR'25**). [link]

[8] Kolouju, P., **Xing, E.**, Pless, R., Jacobs, N., & Stylianou, A. *good4cir: Generating Detailed Synthetic Captions for Composed Image Retrieval.* The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (**CVPRW'25**). [link]

[9] Leão, A., Banda, B., **Xing, E.**, Gudapati, S., Ahmad, A., Lin, J., Sastry, S., Jacobs, N. & Reis, R. *Applications of Artificial Intelligence in Public Health: Analyzing Built Environment and Addressing Spatial Inequities.* Journal of Public Health. [link]

[10] Khanal, S., **Xing, E.**, Dhakal, A., Sastry, S., Xiong, Z., Ahmad, A., & Jacobs, N. *Learning Probabilistic Embeddings for Multi-scale Zero-shot Soundscape Mapping.* The 32nd ACM Multimedia Conference (**ACMMM'24**). [link]

[11] **Xing, E.**, Venkatraman, S., Le, T., & Lee, D. ALISON*: Fast Stylometric Authorship Obfuscation.* The 38th Annual AAAI Conference on Artificial Intelligence (**AAAI'24**). [link]

[12] **Xing, E.**, Liu, L., Xing, X., Qu, Y., Jacobs, N., & Liang, G. *Neural Network Decision-Making Criteria Consistency Analysis via Inputs Sensitivity.* 2022 International Conference on Pattern Recognition (**ICPR'22**), pp. 2328-2334. [link]

[13] **Xing, E.** & Haleem, K. *Motorcycle Safety Investigation in Kentucky Using Machine and Deep Learning Techniques.* 2022 ASCE International Conference on Transportation and Development, pp. 68-80. [link]

[14] **Xing, E.** & Xing, G. *A Toolkit for Assessments in Introductory Programming Courses.* Proceedings of the 54th ACM Technical Symposium on Computer Science Education (SIGCSE'23), p. 1285. [link]

[15] Qu, Y., Yan, D., **Xing, E.**, Zheng, F., Zhang, J., Liu, L., & Liang, G. *Beware the Black-Box of Medical Image Generation: an Uncertainty Analysis by the Learned Feature Space.* 2022 International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'22), pp. 3849-3853. [link]

Experience

Multimodal Vision Research Laboratory Aug 2023 – Present

- Led collaboration across 4 universities with the National Center for Missing and Exploited Children to develop image search systems to support analyst investigations.
- Mentored BS/MS students in numerous projects across representation learning, reinforcement learning, and applied collaborations.
- Led collaboration with WashU Public Health in extracting built environment features with vision-language models to measure spatial inequities.

Pinterest Sep 2025 – Dec 2025

- Trained foundational models for multimodal search as an ML intern with the Advanced Technologies Group.
- Developed training, evaluation, and deployment infrastructure on massive-scale data and compute.
- Proposed joint hypernetwork-encoder architecture for performant and scalable search across Pinterest's image indices.

Student Researcher @ Penn State May 2022 – Aug 2023

- Proposed the novel authorship obfuscation system ALISON based on greedy replacement of important part-of-speech sequences.
- Demonstrated that ALISON achieves 15% greater attack success while maintaining a higher degree of semantic preservation between original and obfuscated texts, all while reducing runtime by $> 10\times$.
- Performed label entropy, parameter, and interpretability analyses on ALISON.

Awards and Service

McKelvey School of Engineering CSE Outstanding Scholar Award

Computing Research Association Outstanding Undergraduate Researcher – Honorable Mention

Reviewer: NeurIPS, CVPR, ICCV, ECCV, AAAI, WACV, ACMMM

Skills

Languages: Python, Java, C/C++, JavaScript, SQL, HTML/CSS, MATLAB

Libraries: PyTorch, PyTorch Lightning, Ray, Optuna, HuggingFace, AWS, TensorFlow, NumPy, Pandas, Scikit-Learn, Stanza, NLTK, Captum, Matplotlib

Additional

Major Coursework: Advances in Computer Vision, Nonlinear Optimization, Bayesian Machine Learning, Reinforcement Learning, Text Mining, Machine Learning, Advanced Algorithms, Automata Theory and Compilers, Data Structures and Algorithm Analysis

Other Coursework: Numerical Analysis, Multivariable Calculus, Discrete Mathematics, Linear Algebra

GRE: 336 *(170 Quantitative, 166 Verbal)*, **5.5** Analytical Writing

US Citizen